

Quiz Result

B2 ADS Assignment

Subject: ADS

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Performance Summary

Your Score: 25 / 40

Percentage: 63%

Correct Answers: 14

Wrong Answers: 6

Questions Attempted: 20



63%

Score Breakdown

Raw Score (correct answers): +28

Total Penalty (wrong answers): -3

Final Score: 28 - 3 = 25

Difficulty Breakdown

Easy Questions Attempted: 0

Hard Questions Attempted: 20

Question-wise Breakdown

#	Question	Your Answer	Status	Score
1	Which of the following is a key advantage of using cloud-based data science platforms over on-premise solutions, particularly for large-scale projects?	A. Scalability and elasticity of resources	Correct	+2
2	A researcher conducts a One-Way ANOVA comparing the effectiveness of three different teaching methods on student test scores. The ANOVA yields a significant F-statistic ($p < 0.05$). Which of the following statements most accurately describes the subsequent interpretative steps?	B. The ANOVA result suggests that a t-test should now be performed between each pair of groups to confirm individual differences, without correction for multiple comparisons.	Wrong	-0.5
3	How do SHAP and LIME primarily differ in their approach to generating model explanations?	C. SHAP is based on game theory for feature contributions, while LIME creates local surrogate models.	Correct	+2
4	A social scientist performs a Chi-Square test for independence on a 3x4 contingency table (3 rows, 4 columns) with a total sample size of $N=500$. The test yields a p-value of 0.008. Which of the following is the most accurate interpretation of this result in isolation?	C. There is a statistically significant association between the row and column variables, and the strength of this association is demonstrably large.	Wrong	-0.5
5	A primary goal of AutoML is to automate which specific aspect of the machine learning pipeline, often enabling non-experts to build models?	C. Model selection, hyperparameter tuning, and ensemble methods	Correct	+2
6	A model trained to predict house prices performs poorly because the relationship between features (e.g., number of rooms) and the target variable (price) has changed over time due to economic shifts. This scenario is an example of:	D. Feature Engineering Error	Wrong	-0.5
7	Which of the following is a common practice to avoid misleading visualizations?	B. Using inconsistent color palettes across different charts for the same metric.	Wrong	-0.5
8	A visualization showing average income by race and gender, where a specific racial group appears to have significantly lower income due to an underrepresentation of high-income individuals in the sample, but this bias is not disclosed, represents a failure in:	C. Transparency and fairness in visualization	Correct	+2
9	Which distribution would be most appropriate for modeling the number of customer arrivals at a store in a given hour?	A. Poisson distribution	Correct	+2
10	Which stage of the data science process typically involves feature engineering, data cleaning, handling missing values, and transforming variables?	D. Data Preparation and Exploration	Correct	+2
11	Which method is generally more robust for detecting outliers in a univariate dataset if the data distribution is heavily skewed?	B. Standard deviation rule	Wrong	-0.5
12	In systematic sampling, if you have a population of 1000 and want a sample of 100, what would be your sampling interval (k)?	B. 10	Correct	+2

#	Question	Your Answer	Status	Score
13	A dataset where measurements consistently produce similar results under the same conditions, but do not accurately reflect the true underlying phenomenon, is considered:	B. High in accuracy, low in precision	Wrong	-0.5
14	Given $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.2$, what is $P(A B)$?	D. 0.4	Correct	+2
15	A company wants to develop a personalized recommendation system for its e-commerce platform. Which data science role would be most critically involved in designing, building, and optimizing the predictive algorithms for this system?	B. Machine Learning Engineer / Data Scientist	Correct	+2
16	What ethical concern is raised when a data visualization intentionally uses a truncated y-axis to make a minor increase appear as a significant surge?	C. Lack of transparency and fairness	Correct	+2
17	In the banking sector, what is a common application of machine learning that helps financial institutions mitigate financial loss by identifying unusual transactions?	D. Fraud detection	Correct	+2
18	What is the primary purpose of data normalization in the data preprocessing phase?	D. To scale numerical features to a standard range	Correct	+2
19	Which of the following best describes the goal of Feature Engineering?	C. To transform raw data into features that better represent the underlying problem to the model	Correct	+2
20	A distribution described as 'leptokurtic' has:	A. Thicker tails and a higher peak than a normal distribution	Correct	+2

Correct Answers for Wrong Responses

Q#	Question	Correct Answer
2	A researcher conducts a One-Way ANOVA comparing the effectiveness of three different teaching methods on student test scores. The ANOVA yields a significant F-statistic ($p < 0.05$). Which of the following statements most accurately describes the subsequent interpretative steps?	D. A significant F-statistic implies that at least two group means are significantly different, but further post-hoc tests are required to identify which specific pairs differ.
4	A social scientist performs a Chi-Square test for independence on a 3x4 contingency table (3 rows, 4 columns) with a total sample size of $N=500$. The test yields a p-value of 0.008. Which of the following is the most accurate interpretation of this result in isolation?	D. The observed cell frequencies across the entire table are unlikely to have occurred by chance if the variables were truly independent.
6	A model trained to predict house prices performs poorly because the relationship between features (e.g., number of rooms) and the target variable (price) has changed over time due to economic shifts. This scenario is an example of:	C. Concept Drift
7	Which of the following is a common practice to avoid misleading visualizations?	C. Always starting the y-axis at zero for bar charts showing absolute values.
11	Which method is generally more robust for detecting	C. Interquartile Range (IQR) method

Q#	Question	Correct Answer
	outliers in a univariate dataset if the data distribution is heavily skewed?	
13	A dataset where measurements consistently produce similar results under the same conditions, but do not accurately reflect the true underlying phenomenon, is considered:	D. High in reliability, low in validity